

Microstructure and compressive properties of AlCrFeCoNi high entropy alloy

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Abstract

An AlCrFeCoNi high entropy alloy was prepared by vacuum arc melting. Only diffraction peak corresponding to a BCC crystal structure is observed for this AlCrFeCoNi high entropy alloy. The microstructure of this AlCrFeCoNi alloy is polygonal grains with intragranular dendritic segregation. Dendritic segregation area is found to be Al, Ni rich and Cr, Fe deplete, while interdendritic segregation area is Cr, Fe rich and Al, Ni deplete. The distribution of Co is essentially identical. The fine microstructure of dendritic segregation area and of interdendritic segregation area is found to be nanoscale spherical precipitates morphology and basket-weave morphology, respectively. Results of EDS attached on high resolution scanning electron microscope (SEM) revealed that these morphological characteristics are also resulted from elements segregation. This AlCrFeCoNi high entropy alloy exhibits excellent compressive properties. The yield stress, compressive strength and plastic strain of the alloy reaches 1250.96, 2004.23 MPa, and 32.7%, respectively. The fracture mechanism of this AlCrFeCoNi high entropy alloy is observed as cleavage fracture and slip separation.

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1. Introduction

Traditional metallurgical theory suggests that multiple alloying elements in an alloy may lead to the formation of many compounds with complex microstructure and poor mechanical properties [1]. Recently this paradigm has been broken by high entropy alloy developed by Yeh et al. [2]. High entropy alloys are defined as alloys that have at least five principal elements and the concentration of each element is between 5 and 35 at.%. Yeh and co-workers [3–5] found that high entropy alloys may possess simple solid solution structure. They considered that the high mixing entropy of multi-principal elements induces the formation of multicomponent solid solution. The high entropy alloys possess excellent properties as well. With proper composition design, promising properties in hardness, wear resistance, oxidation resistance, and corrosion resistance can be obtained [6–9].

These discoveries brought about a new alloy design concept and opened a new research field of metallic materials.

Many high entropy alloy systems have been exploited in the past decade, and most of the reported alloy systems contain Cu [2–7,9]. Cu tends to segregate at grain boundary and improves the plasticity of high entropy alloys [3,10,11]. Chen et al. [11] found the rupture of AlTiFeNiCr_x alloy during cooling, and they considered Cu is the necessary element for high entropy alloy. Zhou et al. [12] studied the room temperature mechanical properties of AlCoCrFeNiTi_x system. They found that the plastic strain of both Ti₀ and Ti_{0.5} alloys exceeded 20%. This indicates that not all the high entropy alloy systems without Cu are brittle. Therefore, in this paper an AlCrFeCoNi high entropy alloy was prepared, and its microstructure and compressive properties were investigated intensively.

2. Materials and experimental details

A high entropy alloy AlCrFeCoNi (nominal composition is equimolar for each element) was prepared using a vacuum arc melting furnace. The purity of all the raw elemental metal is above 99.9%. Melting atmosphere was Ti-gettered high-purity argon. The ingot was remelted four times to improve chemical homogeneity.

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The crystalline structure of alloy was characterized using D/max-rB X-ray diffractometer (XRD). A FEI sirion scanning electron microscope (SEM) was used for microstructure observation. A Hitachi S-5500 scanning electron microscope equipped with EDS was used to identify the element distribution in nanoscale. Room temperature compressive properties was tested on an Instron 5500 testing machine with a loading speed of 0.5 mm/min. The test specimens were cylindrical, $\Phi 3 \text{ mm} \times 5 \text{ mm}$. Three compression tests were performed to obtain the average value.

3. Results and discussion

3.1. Crystal structure determination

The XRD pattern of the AlCrFeCoNi high entropy alloy is shown in Fig. 1. Only diffraction peak corresponding to a BCC crystal structure whose lattice constant is 0.2873 nm is observed. Thus Al, Cr, Fe, Co, Ni elements are expected to distribute in the crystal lattices and form quinary BCC solid solution.

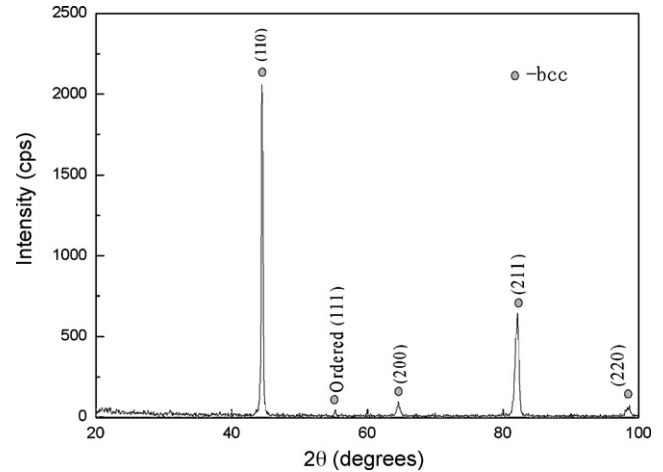


Fig. 1. XRD pattern of AlCrFeCoNi high entropy alloy.

3.2. Microstructure

Fig. 2 shows the SEM images of the as-cast AlCrFeCoNi high entropy alloy. It was found that the produced AlCrFeCoNi

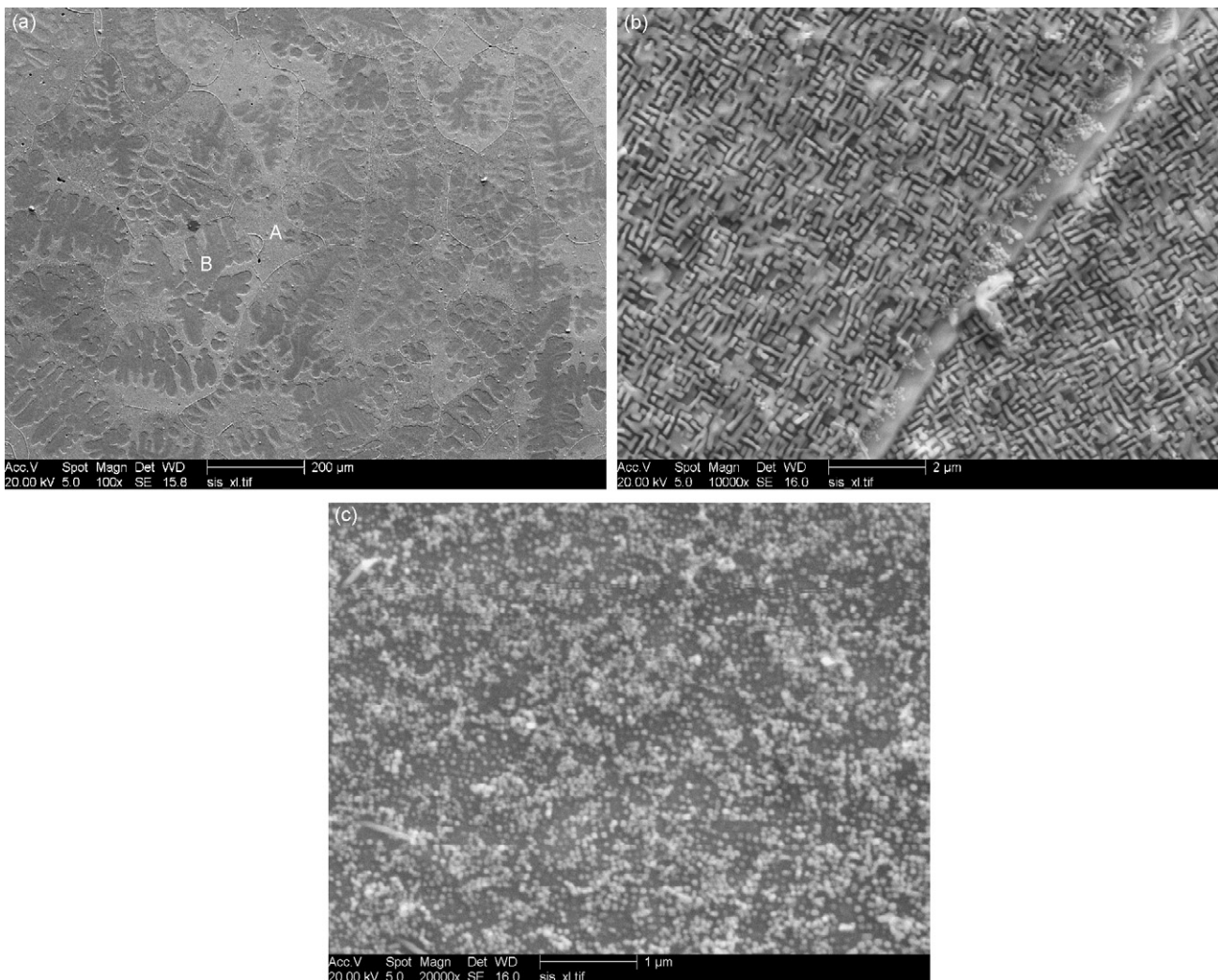


Fig. 2. SEM microstructure of AlCrFeCoNi high entropy alloy. (a) Low magnified image, (b) high magnified image of zone A in (a), (c) high magnified image of zone B in (a).

Table 1
Composition analysis (EDS) results of AlCrFeCoNi high entropy alloy (at.%)

Region	Al	Cr	Fe	Co	Ni
A	18.31	20.39	22.33	20.73	18.24
B	28.77	11.98	16.15	19.24	23.85
S_R	1.57	0.59	0.72	0.93	1.31

alloy is consisted of polygonal grains with intragranular dendritic segregation. The interdendritic segregation area is named zone A and the dendritic area is named zone B.

The chemical compositions of the two zones characterized by EDS are shown in Table 1. A segregation parameter, segregation ratio, S_R , was introduced to demonstrate the degree of segregation of elements, which is defined as follows:

$$S_R = \frac{\text{element concentration in zone B}}{\text{element concentration in zone A}}$$

The calculated results are shown in Table 1 as well. It was clearly shown that Cr and Fe are rich in zone A while Al and Ni are rich in zone B. Only Co does not show evident different in the two zones with a slightly higher concentration in zone A.

Fig. 2(b) is a magnified part of zone A in Fig. 2(a). The fine microstructure of zone A is basket-weave morphology in nanoscale. Fig. 2(c) is a magnified part of zone B in Fig. 2(a). Different from zone A, this area contains many nano-sized spherical precipitates.

More detailed investigations was carried out using high resolution SEM (Hitachi S-5500 scanning electron microscope) equipped with EDS on the nanoscale element distribution. Figs. 3 and 4 shows the fine microstructure of zone A and zone B in Fig. 2(a), respectively. Three types of microstructures were observed in zone A. Zone 1 is straight grain boundary, zone 2 is the basket-weave area and zone 3 is denoted as matrix because its continuity under high resolution observations. EDS analysis (Table 2) indicated that zone 1 and zone 2 are both Cr and Fe rich. However, the segregation degree of zone 1 is weaker than of zone 2. Al and Ni was found rich in zone 3. Co behaved differently with a slightly higher concentration in Cr and Fe rich area.

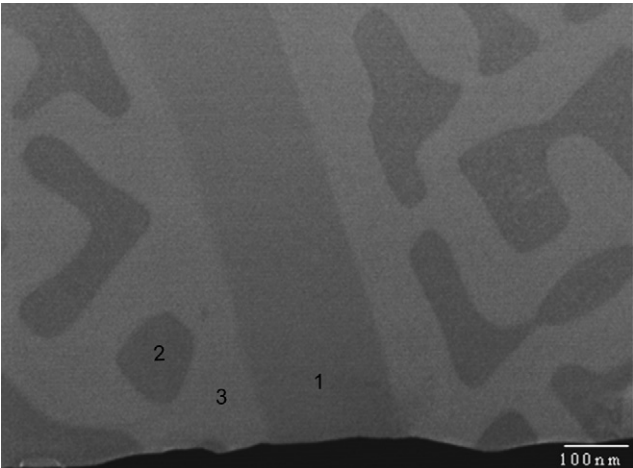


Fig. 3. High resolution SEM microstructure of zone A in Fig. 2(a) of AlCrFeCoNi high entropy alloy.

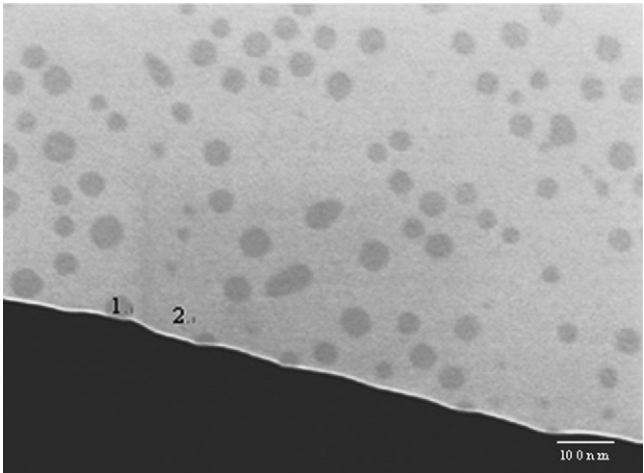


Fig. 4. High resolution SEM microstructure of zone B in Fig. 2(a) AlCrFeCoNi high entropy alloy.

Table 2
Composition analysis (EDS) results of zone A in Fig. 2(a) of AlCrFeCoNi high entropy alloy (at.%)

Region	Al	Cr	Fe	Co	Ni
1	16.86	19.97	23.67	23.20	16.30
2	13.80	20.05	28.42	23.00	14.74
3	23.30	13.29	17.80	20.53	25.07

Table 3
Composition analysis (EDS) results of zone B in Fig. 2(a) of AlCrFeCoNi high entropy alloy (at.%)

Region	Al	Cr	Fe	Co	Ni
1	13.98	30.16	26.00	19.02	10.84
2	23.35	13.34	16.50	21.58	25.24

Fig. 4 shows that zone B is consist of two types of microstructures. Zone 1 is spherical precipitates, and zone 2 is the matrix. The compositions of these two zones were characterized using EDS and summarized in Table 3. The spherical precipitates are Cr and Fe rich, while Al and Ni are rich in matrix. The microscale and nanoscale analysis showed similar element segregation tendency in the alloy, one area with Al and Ni rich, but Cr and Fe depleted and the other Cr and Fe rich, but Al and Ni depleted.

From the mixing enthalpy of atom-pair as shown in Table 4, it clearly shows that the mixing enthalpy of Al and Ni is higher than of other atom-pair, which indicates that Al and Ni atoms tend to form atom pairs and segregate. Heterogeneous atoms

Table 4
Mixing enthalpy of different atom-pair [13]

Element (atomic sizes)	Fe	Cr	Ni	Co	Al
Fe(1.24)	–	–1	–2	–1	–11
Cr(1.25)		–	–7	–4	–10
Ni(1.24)			–	0	–22
Co(1.25)				–	–19
Al(1.43)					–

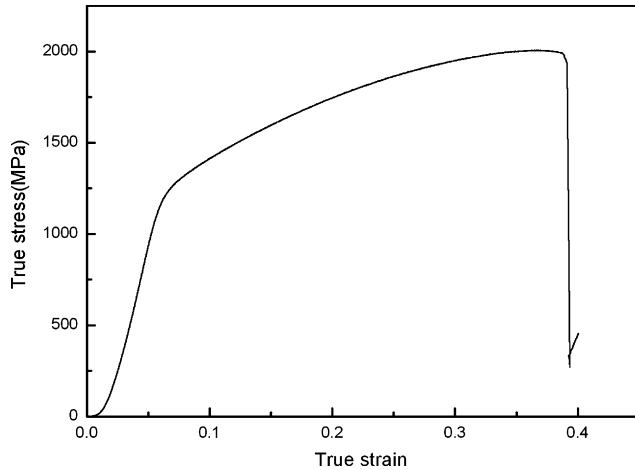


Fig. 5. True stress-true strain curve of AlCrFeCoNi high entropy alloy under compression.

segregation in solid solution form short-range orderly condition. When the temperature is lower, it can form ordering solid solution. The (1 1 1) crystal plane ordered peak shown in Fig. 1 confirm this ordering exist in AlCrFeCoNi high entropy alloy.

3.3. Compressive properties

Fig. 5 is the true stress-true strain curve of AlCrFeCoNi high entropy alloy under compression at room temperature. The yield stress of AlCrFeCoNi alloy is 1250.96 MPa. The compressive strength reaches 2004.23 MPa, with a ultimate plasticity strain is 32.7%. A strong strain hardening ability was observed in AlCrFeCoNi alloy.

Table 5

Room temperature compression properties [14,15]

Alloy	σ_y (MPa)	σ_{max} (MPa)	ϵ_p (%)
AlCrFeCoNi	1250.96	2004.23	32.7
CrFeCoNiCuTi	1272	1272	1.6
AlCrMnFeCoNiCuTiV	1862	2431	0.95
CrFeCoNiCuTi _{0.5}	700	1650	21.26

For substitutional solid solution, heterogeneous atoms with different atomic sizes combined together will lead to lattice distortion. Table 4 lists the atomic sizes of Cr, Fe, Co, Ni and Al, in which the atomic size of Al is much larger. The solution of Al atoms into the crystalline lattice leads to significant increase of lattice distortion energy and, consequently, solid solution strengthening is enhanced.

Table 5 lists the compressive properties of AlCrFeCoNi high entropy alloy and some other high entropy alloys. Although CrFeCoNiCuTi and AlCrMnFeCoNiCuTiV alloys contain Cu, they reveal almost no compression plasticity. CrFeCoNiCuTi_{0.5} alloy is a typical high entropy alloy which possesses high strength and certain plasticity. In contrast, the AlCrFeCoNi alloy without Cu shows better strength and plasticity.

Two typical fracture morphologies were observed for the AlCrFeCoNi high entropy alloy (Fig. 6). Cleavage steps and intergranular fracture were found on the fracture surfaces (Fig. 6(a)). Fig. 6(b) shows the slip lines and shallow dimple-like morphology. The fracture mechanism of AlCrFeCoNi high entropy alloy is therefore concluded to be cleavage fracture and slip separation.

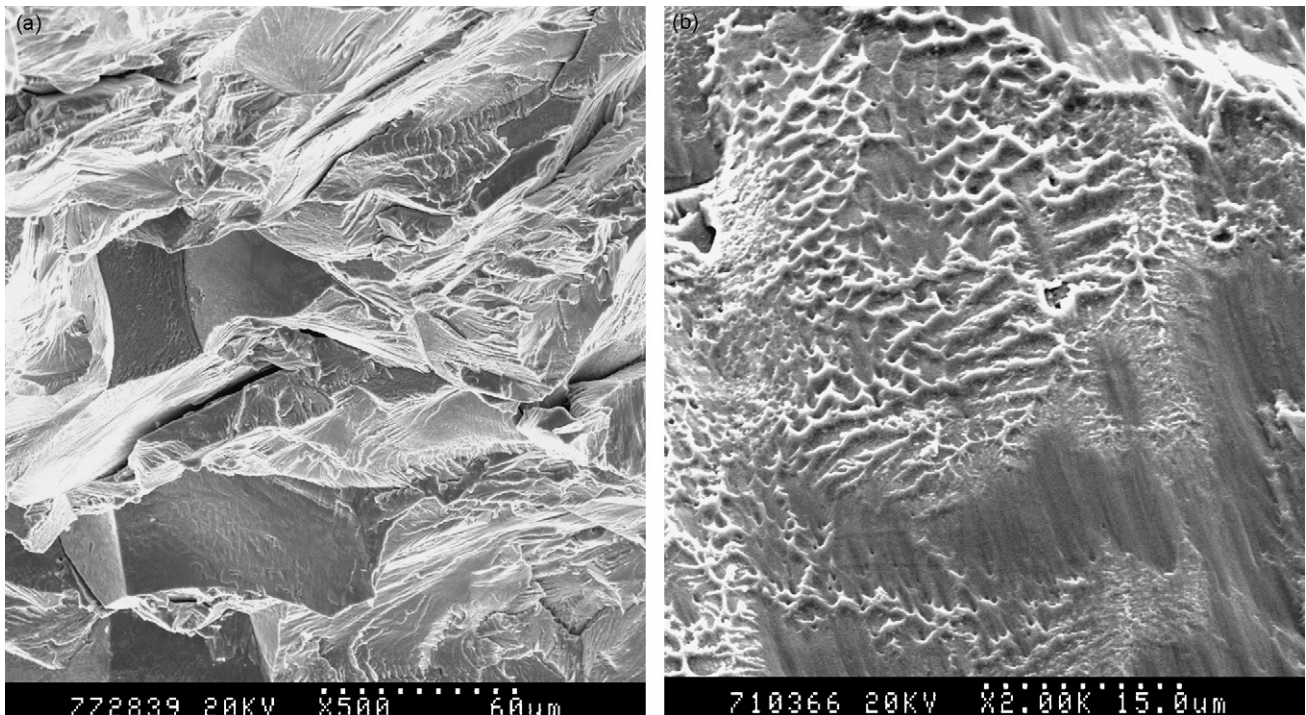


Fig. 6. Fractographic features of AlCrFeCoNi high entropy alloy. (a) Cleavage fracture morphology and (b) slip fracture morphology.

4. Conclusions

An AlCrFeCoNi high entropy alloy was produced with a BCC crystal structure. Its microstructure is polygonal grains with dendritic segregations. The dendritic segregation area is Al and Ni rich but Cr and Fe depleted, while the interdendritic segregation area is Cr and Fe rich but Al and Ni deplete. The distribution of Co does not show evident difference in the two types of microstructure. The fine microstructures of dendritic segregation area and interdendritic segregation area are spherical precipitates and nanoscale basket-weave structure, respectively. Results of EDS attached on high resolution SEM revealed that these morphological characteristics are also resulted from elements segregation. The yield stress of AlCrFeCoNi was measured to be 1250.96 MPa, while the compressive strength reaches 2004.23 MPa. AlCrFeCoNi alloy shows large strain hardening ability and compressive plasticity, 32.7%. The fracture mechanism of AlCrFeCoNi high entropy alloy is proposed to be cleavage fracture and slip separation.

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